

I have one more important concern that should be checked **before continuing with the interpolation, Autoencoder, or BPNN analysis**.

Additional Step: Recheck the Original EIS Data Extraction and Data Cleaning

After looking again at Figures 3, 4, and 6, I am concerned that the problem may already exist at the **raw EIS data extraction or data-cleaning stage**.

The Nyquist plots are unusually irregular, with many sharp changes, reversals, and oscillations. Importantly, this irregularity does not appear only after Cubic Spline interpolation; the raw measured points themselves also appear quite irregular in Figure 4.

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Therefore, before concluding that Cubic Spline, PCHIP, the AE, or the BPNN is causing the problem, please go back to the **original NASA EIS data** and verify the entire EIS extraction process step by step.

Step A. Verify the original NASA impedance variables

Please identify exactly which original variables from the NASA `.mat` files were used to construct:

- $\text{Re}(Z)$
- $\text{Im}(Z)$
- $-\text{Im}(Z)$
- Frequency

Please show the exact mapping from the original NASA data structure to the CSV variables.

For example:

Original NASA variable $\rightarrow$ complex impedance $\rightarrow \text{Re}(Z)$ and $\text{Im}(Z) \rightarrow$ CSV

I want to make sure that the correct impedance variable was extracted and that no sign, scaling, or conversion error occurred during preprocessing.

Step B. Verify the actual frequency vector

Each EIS point should conceptually correspond to:

$$(f_i, \text{Re}(Z_i), \text{Im}(Z_i))$$

Please confirm that the actual measurement frequency is available from the original NASA data.

Do not assume that the CSV row order represents frequency order unless this has been verified from the original dataset.

If the frequency information was lost when creating the CSV, please regenerate the EIS dataset and preserve the frequency column.

The new dataset should preferably contain:

Battery	Cycle	Frequency	Re(Z)	-Im(Z)
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Step C. Verify the ordering of the EIS points

For one representative example, such as B0005 at Cycle 41, please show all raw EIS points ordered by the **actual measured frequency**.

Please provide a table such as:

Point	Frequency	Re(Z)	-Im(Z)
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Then create three separate plots:

1. **Raw measured EIS points only**
2. **Raw points connected according to actual frequency order**
3. **Raw points plus Linear, Cubic, and PCHIP interpolation**

This will allow us to determine whether the irregular Nyquist curve comes from the original measurement itself or from an incorrect point-ordering/preprocessing procedure.

Step D. Check whether the 39 points really belong to the same EIS spectrum

Please also confirm that all 39 points assigned to one battery cycle belong to the same impedance measurement.

We should make sure that the data conversion process did not accidentally combine points from different EIS measurements.

For each spectrum, verify:

- same Battery ID
- same EIS measurement/cycle
- correct number of frequency points
- no mixing between different measurements

Step E. Perform basic EIS data-quality checks

Before any interpolation, please check the raw EIS data for:

- NaN values
- Inf values
- missing frequency points
- duplicated frequencies
- duplicated impedance values
- inconsistent frequency ordering
- abnormal jumps or isolated outliers
- correct units, such as Ω versus $m\Omega$
- correct sign convention for $\text{Im}(Z)$ and $-\text{Im}(Z)$

Please summarize how many samples are affected by each issue.

However, **do not remove or smooth abnormal-looking points only because the Nyquist plot looks unattractive**. Any data-removal rule must have a clear and reproducible justification.

Step F. Do not use smoothing to make the Nyquist curve look better

At this stage, please do not apply smoothing simply to obtain a visually nicer semicircle.

If the frequency order or the extracted variable is incorrect, smoothing would only produce:

a smoother version of an incorrectly processed EIS signal.

The correct order should be:

Original NASA data

- verify original variables
- verify actual frequency
- verify frequency ordering
- verify $\text{Re}(Z)/\text{Im}(Z)$ sign and units
- verify spectrum grouping
- check missing values/outliers
- only then perform interpolation
- then evaluate AE
- then evaluate BPNN/SOH prediction.

Step G. Reconsider the current interpretation of Figures 3 and 4

At present, the report mainly interprets the irregularity as an interpolation problem and concludes that Cubic Spline introduces physically invalid overshoot.

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However, before making that conclusion, we need to distinguish two possibilities:

Case 1: The raw Nyquist curve becomes reasonable after using the correct frequency ordering.

If this happens, then the current preprocessing or ordering is likely incorrect.

Case 2: The correctly ordered original NASA EIS points are still highly irregular.

If this happens, then we need to determine whether the irregularity comes from the original NASA measurement, measurement noise, or the specific impedance variable being used.

Only after this is confirmed should we compare Linear, Cubic, and PCHIP interpolation.

Step H. This issue may also explain the poor SOH result in Figure 10

Figure 10 currently shows very poor generalization: the true capacity decreases with cycling, but the predicted trajectory is almost constant, with a negative R^2 .

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At present, the report attributes this mainly to AE latent-feature instability.

However, we should not yet assume that the AE is the original source of the problem.

There may be an earlier issue in the pipeline:

Incorrect EIS extraction/order → incorrect EIS representation → unstable AE features → poor BPNN prediction

In that case, the observed “correlation flip” would be a consequence of an earlier preprocessing problem rather than the original cause.

Therefore, please verify the EIS source data first before making a strong conclusion about AE-induced domain shift.

Priority for the next update

Before continuing with the larger AE/BPNN ablation experiments, please first complete the following:

1. Identify the original NASA EIS variables.
2. Recover and verify the true measurement-frequency vector.
3. Confirm the frequency order of the EIS points.

4. Confirm that all points belong to the same EIS spectrum.
5. Verify $\text{Re}(Z)$, $\text{Im}(Z)$, sign convention, and units.
6. Check missing values, duplicated points, and potential outliers.
7. Redraw the raw Nyquist plot **without interpolation**.
8. Only after the raw Nyquist curve is verified, repeat the Linear/Cubic/PCHIP comparison.
9. Reconstruct Figure 6 using the corrected EIS preprocessing.
10. Only then rerun the AE-BPNN SOH experiments.

The key point is:

Before optimizing the model, we must first prove that the EIS input itself is correct.

If the EIS extraction or preprocessing is incorrect, then the later AE/BPNN results, including the 108-model ablation study, may need to be repeated.

資料來源